

Chiral Lattice Resonances in 2.5-Dimensional Periodic Arrays with Achiral Unit Cells

Luis Cerdán, Lauren Zundel, and Alejandro Manjavacas*



Cite This: *ACS Photonics* 2023, 10, 1925–1935



Read Online

ACCESS |



Metrics & More



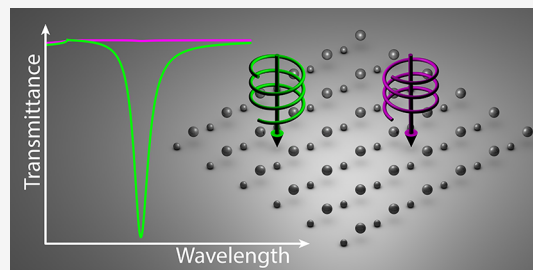
Article Recommendations



Supporting Information

ABSTRACT: Lattice resonances are collective electromagnetic modes supported by periodic arrays of metallic nanostructures. These excitations arise from the coherent multiple scattering between the elements of the array and, thanks to their collective origin, produce very strong and spectrally narrow optical responses. In recent years, there has been significant effort dedicated to characterizing the lattice resonances supported by arrays built from complex unit cells containing multiple nanostructures. Simultaneously, periodic arrays with chiral unit cells, made of either an individual nanostructure with a chiral morphology or a group of nanostructures placed in a chiral arrangement, have been shown to exhibit lattice resonances with different responses to right- and left-handed circularly polarized light. Motivated by this, here, we investigate the lattice resonances supported by square bipartite arrays in which the relative positions of the nanostructures can vary in all three spatial dimensions, effectively functioning as 2.5-dimensional arrays. We find that these systems can support lattice resonances with almost perfect chiral responses and very large quality factors, despite the achirality of the unit cell. Furthermore, we show that the chiral response of the lattice resonances originates from the constructive and destructive interference between the electric and magnetic dipoles induced in the two nanostructures of the unit cell. Our results serve to establish a theoretical framework to describe the optical response of 2.5-dimensional arrays and provide an approach to obtain chiral lattice resonances in periodic arrays with achiral unit cells.

KEYWORDS: Chirality, Lattice resonances, Periodic arrays, Nanoparticle arrays, 2.5-Dimension, Complex unit cell



INTRODUCTION

Periodic arrays of metallic nanostructures support collective electromagnetic modes commonly known as lattice resonances.^{1–3} These modes, which appear in the spectrum at wavelengths commensurate with the periodicity of the array, are the result of the coherent multiple scattering between the individual constituents.^{4,5} Due to their collective nature, lattice resonances give rise to very narrow optical responses,^{6–9} with extraordinarily high quality factors for systems made of metallic nanostructures,^{10–13} and couple strongly with light,^{14–16} producing values of reflectance and absorbance that can reach the theoretical limits for two-dimensional systems.^{17,18} Such exceptional properties have made lattice resonances the subject of an extensive research effort, which has resulted in the proposal and development of a wide range of applications. These include, among others, ultrasensitive biosensors,^{19–21} color filters,^{22,23} light-emitting devices,^{24–28} light-to-heat transducers,^{29,30} and even platforms to explore new physical phenomena.^{31–36}

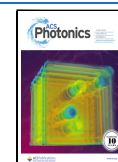
More recently, lattice resonances have begun to be explored for applications involving chirality.^{37–40} A material object is said to be chiral when it cannot be superimposed on its mirror image, or enantiomer, by rotations and translations.⁴¹ Many important biomolecules, such as sugars and amino acids, as well as drugs used in medicine, are chiral^{42,43} and, in many

cases, only one enantiomer is active while the other is inactive or even harmful.^{44,45} Therefore, the precise enantioselective detection, identification, and manipulation of chiral molecules with light-based technologies is a key goal in biosensing and photochemistry, which requires the development of optical systems with strong chiral responses.^{46–52} These systems are also very relevant for the implementation of optical elements capable of controlling and modifying the polarization state of light.^{53–55}

The prospects of these applications have triggered the study and characterization of a variety of periodic arrays supporting chiral lattice resonances.^{56–64} The majority of these arrays are built from the repetition of chiral unit cells, made of either individual nanostructures with complex chiral morphologies,^{56–58,63} such as helices,⁶⁰ or multiple nanostructures forming a chiral arrangement.^{65,66} As expected, the lattice resonances supported by these systems exhibit different responses under illumination with right- and left-handed

Received: March 17, 2023

Published: May 10, 2023



circularly polarized light. More recently, it has been shown that the combination of a rectangular lattice with an achiral unit cell results in arrays capable of supporting lattice resonances with some degree of chirality, which can be enhanced by positioning the array in an inhomogeneous dielectric environment.⁵⁹ Other alternatives that have been explored to obtain chiral lattice resonances in arrays with achiral unit cells involve the use of higher-order lattice resonances⁶⁷ or the introduction of extrinsic chirality^{68,69} through the excitation of the system with tilted illumination.^{70–72}

Motivated by these developments, here, we present a different approach to obtain chiral lattice resonances in periodic arrays with achiral unit cells, which is based on the use of 2.5-dimensional arrays. Using a rigorous coupled dipole model, we investigate the lattice resonances supported by square bipartite arrays made of unit cells in which the metallic nanostructures are placed at arbitrary positions in all three spatial dimensions. We label these systems as 2.5-dimensional arrays due to the combination of the two-dimensional nature of the lattice and the three-dimensional geometry of the unit cell. Through a comprehensive analysis of their optical response, we show that these systems can support lattice resonances with almost perfect chiral responses and quality factors exceeding 10^4 for normal incidence excitation, despite being made of completely achiral unit cells containing two spherical nanostructures. We explain this behavior as the result of the constructive and destructive interference between the different components of the electric and magnetic dipoles induced in the nanostructures of the unit cell. The results of this work set the foundations to investigate the optical response of 2.5-dimensional arrays and provide the fundamental insight necessary to achieve lattice resonances with strong chiral responses using periodic arrays made of achiral unit cells.

RESULTS AND DISCUSSION

The system under consideration is depicted in Figure 1. It consists of an infinite square array of period a with a unit cell

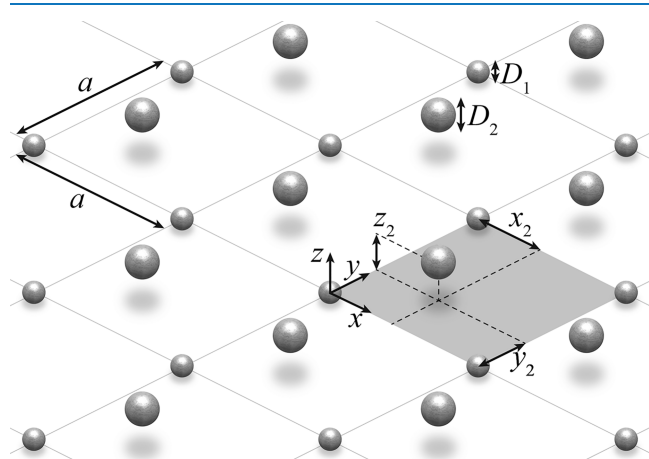


Figure 1. Schematics of the array under study, built from the periodic repetition of a unit cell containing two silver nanospheres (shaded area) over a square lattice with period a . The nanospheres have diameters D_1 and D_2 , respectively, and are located at positions $\mathbf{r}_1 = (0, 0, 0)$ and $\mathbf{r}_2 = (x_2, y_2, z_2)$ with respect to the origin of the unit cell. The array is placed in the xy -plane, surrounded by vacuum, and excited by a circularly polarized electromagnetic plane wave with wavelength λ that propagates along the z -axis.

composed of two nanospheres. Particle 1 has diameter D_1 and is located at $\mathbf{r}_1 = (0, 0, 0)$, i.e., the origin of the unit cell, while particle 2 has diameter D_2 and is placed at $\mathbf{r}_2 = (x_2, y_2, z_2)$. The array lies in the xy -plane and is surrounded by vacuum. As explained above, we refer to this array as a 2.5-dimensional system. Unless otherwise stated, we assume arrays made of silver nanoparticles with $a = 800$ nm, $D_1 = 120$ nm, and $D_2 = 190$ nm throughout the manuscript. We describe the dielectric function of silver using the tabulated data compiled in ref 73. To investigate the chiral properties of the optical response of the array, we excite it with either right-handed circularly polarized (RCP) or left-handed circularly polarized (LCP) electromagnetic plane waves with wavelength λ that propagate along the z -axis. Notice that we use Gaussian units and define the handedness of the circularly polarized light such that the electric $\mathbf{E}$ and magnetic $\mathbf{H}$ fields satisfy $\mathbf{H} = -i\mathbf{E}$ for RCP and $\mathbf{H} = i\mathbf{E}$ for LCP.

It is important to remark that, in contrast to previous works, we consider a square array to maximize the symmetry of the unit cell.⁵⁹ Furthermore, a unit cell composed of two spherical nanoparticles is both intrinsically and extrinsically achiral regardless of their relative position and size.⁷⁴ This can be understood by realizing that the two nanospheres, together with the wavevector of the electromagnetic plane wave that excites them, define a plane. Therefore, the mirror image of such a system can always be superimposed to the original by performing a rotation.⁷⁵ Accordingly, any chiral response displayed by the 2.5-dimensional array under consideration must arise from the collective electromagnetic interactions between its constituents.

We are interested in the optical response associated with the lowest-order lattice resonance of the array, which appears in the spectrum at wavelengths slightly larger than the first Rayleigh anomaly, i.e., at $\lambda \gtrsim a$. Importantly, the size of the nanoparticles forming the arrays under consideration are significantly smaller than a . Consequently, their localized plasmons are spectrally located far from the lattice resonances, thus ensuring that the latter are highly collective.^{30,76} Furthermore, under these conditions, we can describe the optical response of the array using the well-established coupled dipole model (CDM).^{4,5,9,77–80} Within this semianalytical approach, each nanoparticle is modeled as a point dipole with both electric and magnetic components, and the collective response of the array arises from the self-consistent mutual interaction between the dipoles. To capture the rich phenomenology expected for the arrays under study, we need to take into account the magnetic response of the particles^{36,79} as well as the multipartite and three-dimensional nature of the unit cell.^{4,9,13,81} The resulting approach, which is explained in detail in the Methods, is in excellent agreement with full numerical solutions of Maxwell's equations obtained using a finite element method (FEM) solver, as demonstrated in Figure S1 of the Supporting Information. However, in contrast to the full numerical solutions, the CDM provides the necessary insight to understand the origin of the optical response of the array.

Once the induced dipoles are computed, we calculate the transmittance T , reflectance R , and absorbance A of the array from the balance between the power that it radiates and absorbs and the power that is incident on it (see the Methods). Moreover, to characterize the chiral response of the array, we use the dissymmetry factor, defined as^{59,60}

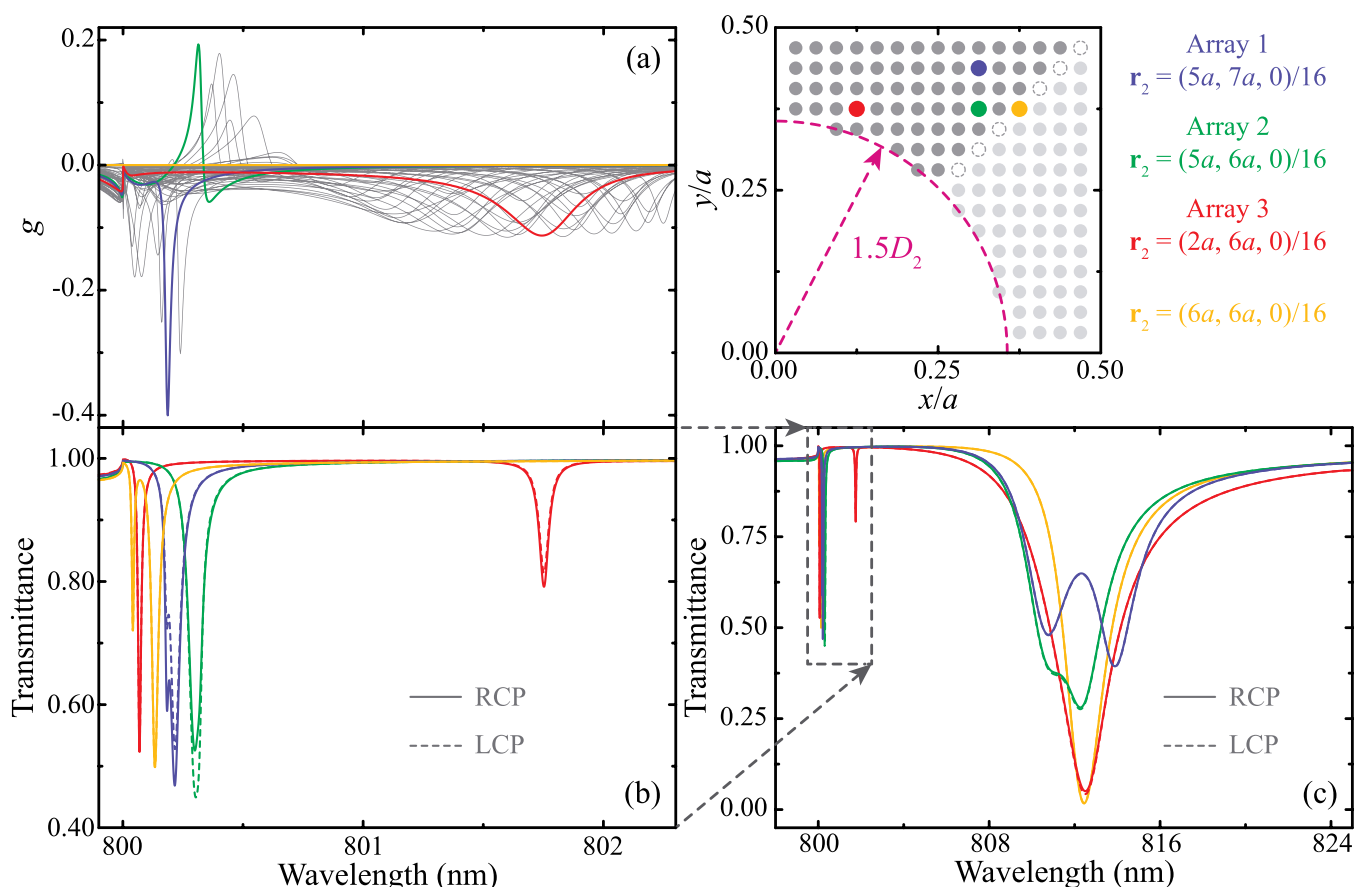


Figure 2. Chiral response of planar arrays. (a) Spectra of the dissymmetry factor for two-dimensional square bipartite arrays in which particle 2 is located at the different positions marked in the accompanying diagram of the lower left quadrant of the unit cell. (b) Spectra of the transmittance under RCP (solid lines) and LCP (dashed lines) excitation for the four selected positions. (c) Zoom-out of panel (b). The color code in all panels matches the one used in the diagram of the quadrant of the unit cell. For all of the arrays, $a = 800$ nm, $D_1 = 120$ nm, and $D_2 = 190$ nm.

$$g = \frac{T_{\text{RCP}} - T_{\text{LCP}}}{1 - \bar{T}}$$

where T_{RCP} and T_{LCP} are the transmittances for RCP and LCP excitation, respectively, and $\bar{T}$ is their average. This quantity, which is inspired by Kuhn's dissymmetry factor (or g -factor) used in spectroscopy,⁸² quantifies the circular dichroism of the array. Notice that, by construction, $-2 \leq g \leq 2$, while the choice of sign is arbitrary and bears no physical meaning.

We begin our study by analyzing the chiral response of conventional two-dimensional square bipartite arrays in which particle 2 is located in the xy -plane. Figure 2(a) shows the value of the dissymmetry factor g as a function of wavelength for different positions of particle 2 within the unit cell. Specifically, we consider the positions signaled with blue, green, red, and dark gray dots in the diagram located to the right of Figure 2(a). These positions span the upper bisector of the lower left quadrant of the unit cell and fulfill the condition $|\mathbf{r}_1 - \mathbf{r}_2| > 1.5D_2$, so that the dipole approximation is sufficiently accurate. Furthermore, given the square symmetry of the array, they are representative of the entire unit cell. For instance, the positions marked with light gray dots are equivalent to those under consideration upon a rotation of 180° around the diagonal of the unit cell. Such rotation is tantamount to changing the handedness of the incident light and, hence, each position below the diagonal results in the same g but with opposite sign as the equivalent position above it. The same argument holds true for the positions $(x_2, a - y_2)$

and $(a - x_2, y_2)$ in the other quadrants, which are equivalent to (x_2, y_2) upon a rotation of 180° around the x -axis and y -axis, respectively. However, positions $(a - x_2, a - y_2)$ result in the same value and sign of g as (x_2, y_2) because they are connected by a 180° rotation around the z -axis.

Examining the results displayed in Figure 2(a), we observe that all of the cases under consideration show a significant chiral response despite corresponding to a two-dimensional square array with an achiral unit cell. The resonant features of g in the different spectra can be roughly distributed in two different regions: a first group is located at wavelengths right above the first Rayleigh anomaly, i.e., $\lambda \gtrsim a = 800$ nm, and a second one at wavelengths around $\lambda \sim 801.5$ nm. To deepen our analysis, we select three examples representative of these groups, which are indicated with blue, green, and red curves, and correspond, respectively, to the positions $\mathbf{r}_2 = (5a, 7a, 0)/16$, $\mathbf{r}_2 = (5a, 6a, 0)/16$, and $\mathbf{r}_2 = (2a, 6a, 0)/16$. For the sake of simplicity, we refer to these three examples as Arrays 1, 2, and 3 (see diagram). Analyzing the transmittance spectra for these arrays, which are shown in Figure 2(b) for RCP (solid curves) and LCP (dashed curves) excitation, we observe that each feature in the spectrum of g corresponds to a lattice resonance. However, the converse is not true. For instance, Array 3 (red curve) supports two lattice resonances in the spectral region under consideration, but only one of them shows a significant value of g . Overall, the lattice resonances have a relatively moderate strength, with values of transmittance around 0.5,

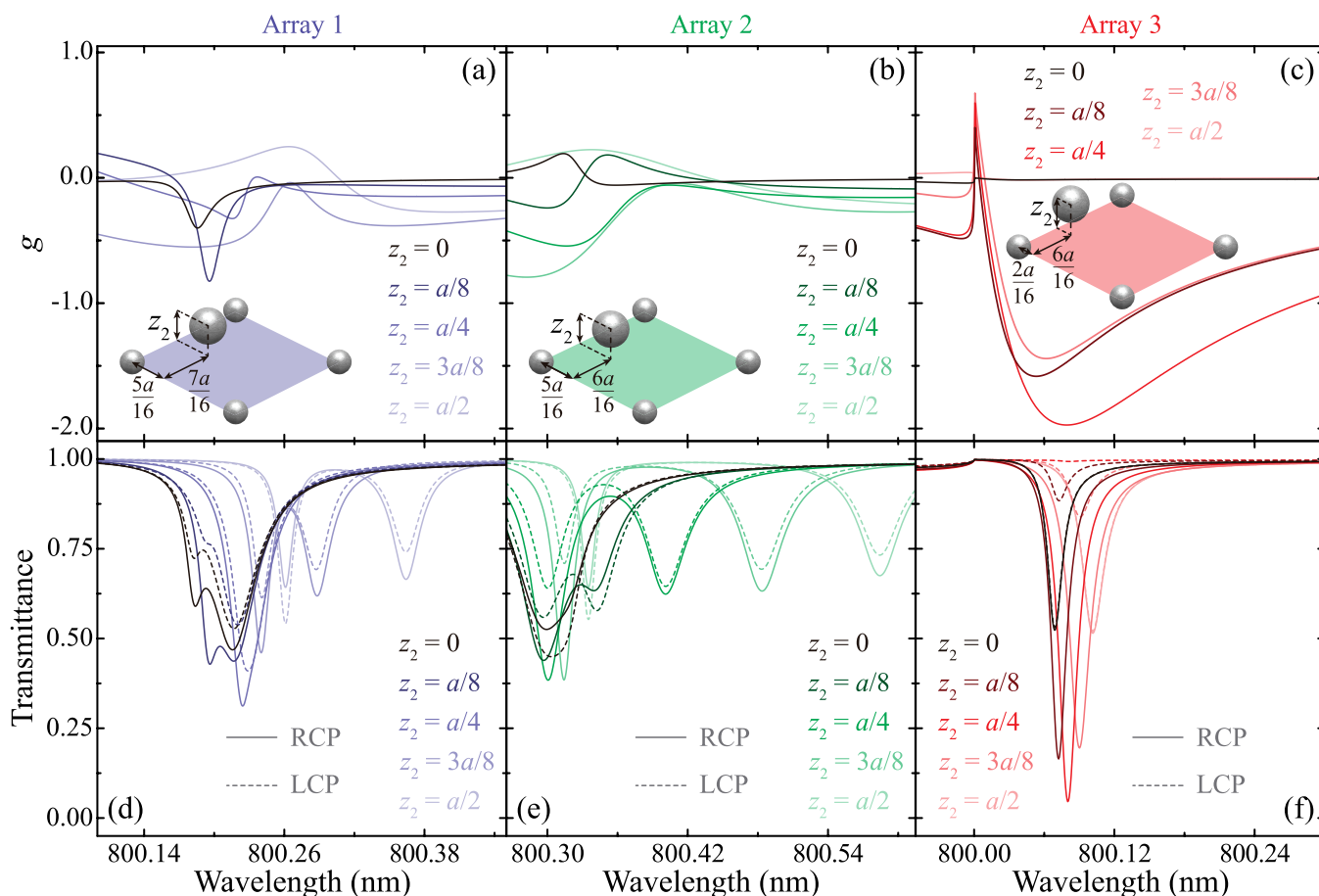


Figure 3. Chiral response of 2.5-dimensional arrays. Dissymmetry factor (a–c) and transmittance (d–f) spectra for three arrays of Figure 2(b) with particle 2 located at different heights z_2 above the xy -plane (see legend). Specifically, we consider Array 1 with $\mathbf{r}_2 = (5a, 7a, z_2)/16$ (a,d), Array 2 with $\mathbf{r}_2 = (5a, 6a, z_2)/16$ (b,e), and Array 3 with $\mathbf{r}_2 = (2a, 6a, z_2)/16$ (c,f). In all panels, solid and dashed curves indicate RCP and LCP excitation, respectively.

and a quality factor of $\sim 10^4$. Importantly, all of these arrays support a much stronger and broader lattice resonance at larger wavelengths, as can be seen in the zoomed-out spectra shown in Figure 2(c). However, these modes display a negligible chirality, as demonstrated by the practically identical transmittance spectra obtained under RCP and LCP excitation.

To complete the analysis of the two-dimensional arrays, we also calculate the spectrum of g and T for an array in which particle 2 is located along the diagonal of the unit cell at $\mathbf{r}_2 = (6a, 6a, 0)/16$ (see the yellow dot in the diagram). The corresponding results, which are displayed in Figures 2(a)–(c) with a yellow curve, show that, in this case, the array supports several lattice resonances in the spectral regions under consideration, but, as expected from the highly symmetric position of the particles in the unit cell, they have no chiral response. The same results are obtained for other positions located along any of the remaining symmetry axes of the unit cell.

With this analysis, we have shown that planar bipartite arrays can display a chiral response if the particles in the unit cell are arranged in low-symmetry configurations. However, the lattice resonances supported by these systems only show a moderate level of strength and chirality. A potential way of improving the response of these modes is to further decrease the symmetry of the unit cell by placing particle 2 outside of the xy -plane. The resulting array, which has no mirror symmetry with respect to

the xy -plane, can be considered a 2.5-dimensional system. We explore this strategy in Figure 3, where we analyze the optical response of 2.5-dimensional arrays in which particle 2 is located at different heights z_2 above the xy -plane. In particular, we focus on the three representative arrays highlighted in Figure 2(b). These are Array 1 with $\mathbf{r}_2 = (5a, 7a, z_2)/16$, Array 2 with $\mathbf{r}_2 = (5a, 6a, z_2)/16$, and Array 3 with $\mathbf{r}_2 = (2a, 6a, z_2)/16$. Figure 3(a) shows the evolution with z_2 of the dissymmetry factor of Array 1. As anticipated, moving particle 2 out of plane leads to a stronger chiral response, with about a twofold increase of g when the particle is displaced from $z_2 = 0$ to $z_2 = a/8$. However, a further increase of z_2 up to $a/2$ worsens the chiral response, suggesting that there is an optimal height. In addition, the feature in the spectrum of g changes its shape, transitioning from a clean Lorentz-like lineshape for $z_2 = 0$ to more complex and broader lineshapes for $z_2 \neq 0$. Array 2, analyzed in Figure 3(b), shows very similar trends, although, in this case, the largest values of g are obtained for $z_2 = 3a/8$. These behaviors can be understood by examining the corresponding transmittance spectra, which are plotted in Figures 3(d) and (e) with solid and dashed curves, respectively, for RCP and LCP excitation. In particular, we observe that, for $z_2 = 0$, both arrays display a pair of overlapping lattice resonances whose separation grows as z_2 is increased. As this happens, the relative values of the transmittance of each of these resonances under RCP and

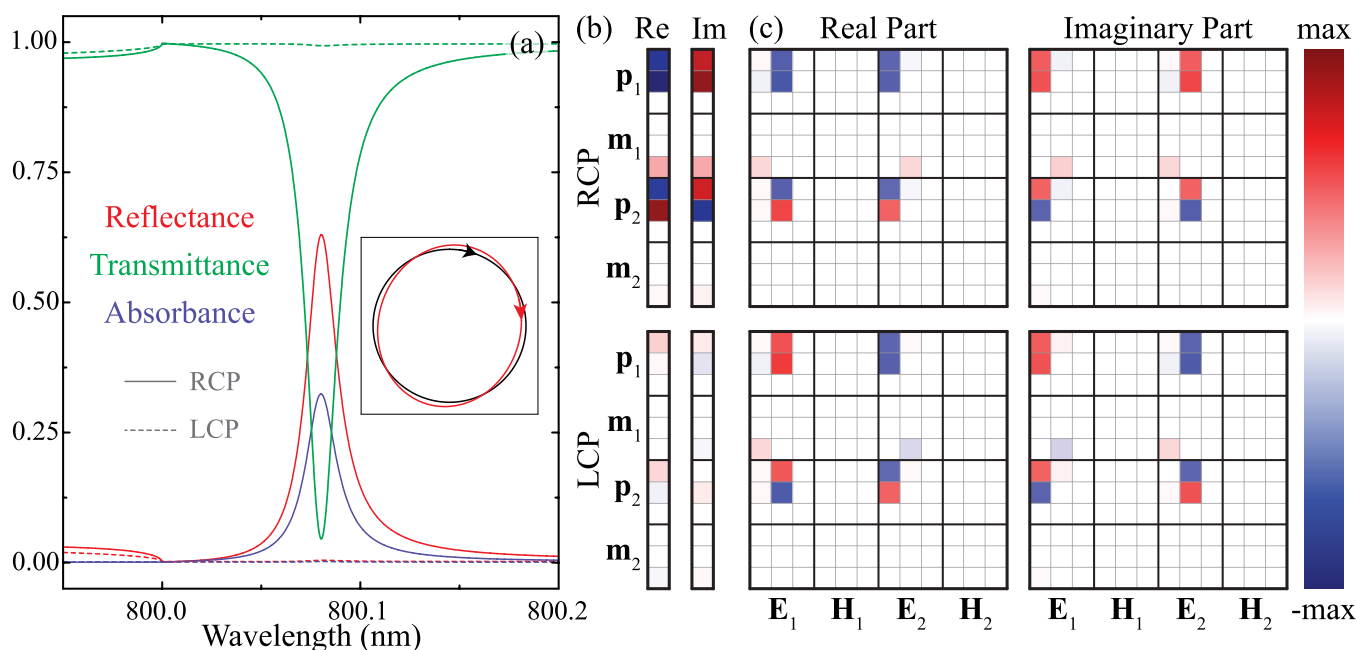


Figure 4. Analysis of the origin of the chiral response. (a) Reflectance, absorbance, and transmittance spectra of the 2.5-dimensional array with $a = 800$ nm, $D_1 = 120$ nm, $D_2 = 190$ nm, and $r_2 = (2a, 6a, 4a)/16$. We use solid and dashed curves to indicate RCP and LCP excitation. The inset shows the polarization state of the incident (black curve) and reflected (red curve) field for RCP excitation, calculated at the lattice resonance wavelength ($\lambda = 800.08$ nm). (b) Real and imaginary parts of the electric and magnetic components of the dipole moments induced in the two particles of the unit cell at the lattice resonance wavelength. (c) Real and imaginary parts of the decomposition of the dipole moments of (b) (see main text). In both cases, the top (bottom) panels correspond to RCP (LCP) excitation. Notice that we normalize all of the quantities shown in panels (b) and (c) to the amplitude of the corresponding incident field, and, therefore, all of them have units of volume. The color bar is common for panels (b) and (c), and its maximum value is 4.1×10^6 nm³.

LCP excitation also change, which, in turn, results in the evolution of the lineshape for g discussed above. Importantly, the strength of the lattice resonances supported by these two arrays does not improve significantly with respect to the values obtained for $z_2 = 0$.

The situation is significantly different for Array 3, whose response is analyzed in Figures 3(c) and (f). For $z_2 = 0$, although this array supports two lattice resonances within the spectral range considered in Figure 2(b), only the resonance located at $\lambda \approx 801.8$ nm displays a chiral response. However, when z_2 increases, the resonance closer to the Rayleigh anomaly also begins to display a chiral response, which becomes maximum for $z_2 = a/4$. For that configuration, the dissymmetry factor reaches a value of $g = -1.97$, i.e., nearly a perfect chiral response. Simultaneously, the transmittance of the array for RCP shows a pronounced dip with a value that drops to $T \sim 0.05$ at the lattice resonance, while, for LCP, it remains at $T \sim 0.99$. These results demonstrate that, if properly designed, 2.5-dimensional arrays can support lattice resonances with maximum chiral response. Importantly, as opposed to some previous works, here, the largest value of g is reached at the wavelength at which the optical response of the array is the strongest: the peak of the lattice resonance.

As an additional comment, we note that, for the three arrays analyzed in Figure 3, after reaching its maximum value, the value of g decreases as z_2 approaches $a/2$. Although it is not shown in Figure 3, increasing z_2 beyond $a/2$ results, first, in a moderate increase of g but with a change of its sign, and then in a gradual decrease until it eventually becomes negligible. Furthermore, moving particle 2 in the opposite direction, that is, along the negative z -axis, leads to the exact same results discussed above, but with the opposite sign of g .

So far, we have focused on describing the chiral lattice resonances supported by 2.5-dimensional bipartite arrays, but it is equally important to understand their physical origin. To that end, we center our attention on Array 3 with $z_2 = a/4$, i.e., $r_2 = (2a, 6a, 4a)/16$, since this system supports the strongest lattice resonance with the largest chiral response among the arrays analyzed. We plot the transmittance, reflectance, and absorbance spectra of this array in Figure 4(a), for both RCP (solid curves) and LCP (dashed curves) excitation. Examining these results, we observe that, for RCP excitation, the strong dip in the transmittance associated with the lattice resonance corresponds to a peak in both the reflectance and the absorbance, with values at resonance ($\lambda = 800.08$ nm) of $T \approx 0.05$, $R \approx 0.63$, and $A \approx 0.32$, and a quality factor of 4×10^4 . In sharp contrast, the array shows an almost negligible response under LCP excitation, resulting in $T \approx 0.99$ at that same wavelength. Therefore, this system behaves as a practically perfect circular dichroic filter, which blocks 95% of RCP light but transmits practically all LCP light. Moreover, it also acts as a chiral mirror that reflects only RCP light with 63% efficiency. Unlike conventional mirrors, which flip the handedness of circularly polarized light after reflection, chiral mirrors preserve it.⁸³ To verify this behavior, we plot, in the inset of Figure 4(a), the polarization state of the incident (black curve) and reflected (red curve) field for RCP excitation. We calculate the polarization state at the wavelength of the lattice resonance taking into account the propagation direction. As anticipated, the field reflected by the array upon RCP excitation has the exact same polarization as the incident one, with only a slight change in ellipticity. Importantly, the same behavior can be obtained for LCP excitation by changing the sign of z_2 .

The chiral lattice resonance supported by the 2.5-dimensional array is the result of the collective interactions between its constituents. Therefore, we can get a deeper insight into their physical origin by looking at the dipole induced in the particles of the array. To that end, in Figure 4(b), we plot the electric and magnetic components of the dipoles induced in particles 1 and 2 of the array considered in Figure 4(a) at the wavelength of the lattice resonance. As one might have expected, the induced dipoles are much stronger for RCP (top panels) than for LCP (bottom panels) excitation. Furthermore, the main contributions to these dipoles are the x and y components for the electric dipoles and the z component for the magnetic ones. Since we are always considering normal incidence, the excitation of a magnetic dipole along the z -axis is a direct consequence of the interaction between the particles. Comparing the magnitude of the electric and magnetic dipoles, we observe that the latter are significantly smaller. Despite this, the magnetic response of the particles plays a crucial role. As shown in Figure S2 of the Supporting Information, if the magnetic response of the particles is neglected, the chirality of the lattice resonance completely disappears.

To understand how the interactions between the particles result in these induced dipoles, we can dissect the different contributions that give rise to them. As explained in Equation 2 of the Methods, the dipoles induced in the particles $\mathbf{d}_i = (\mathbf{p}_i, \mathbf{m}_i)$ are the result of $\mathbf{d}_i = \sum_{j=1}^2 \mathcal{A}_{ij} \mathbf{F}_j$, where $\mathcal{A}_{ij}$ is the ij -th block of the effective polarizability of the array $\mathcal{A}$, and $\mathbf{F}_j = (\mathbf{E}_j, \mathbf{H}_j)$ are the components of the incident electromagnetic field acting on particle j of the unit cell. Notice that, since we use Gaussian units, $\mathbf{p}_i$ and $\mathbf{m}_i$ have the same units, as do $\mathbf{E}_j$ and $\mathbf{H}_j$. In Figure 4(c), we plot the real and imaginary parts of each of the components resulting from the element-wise multiplication of each row of $\mathcal{A}$ with $\mathbf{F}$. The top and bottom panels display the results for RCP and LCP excitation, respectively. The sum of the elements in each row results in the component of the induced dipole located immediately to their left. Interestingly, both polarizations excite the exact same components, namely, the xx , xy , yx , and yy for the electric dipoles, and zx and zy for the magnetic ones. The crucial difference is that, while, for RCP excitation, the terms with $i = j$ have the same sign as those with $i \neq j$, and hence add up constructively, for LCP excitation, these terms have opposite signs, thus canceling each other. Therefore, based on these results, we can conclude that the origin of the chiral lattice resonances supported by 2.5-dimensional bipartite arrays is the constructive and destructive interference between the electric and magnetic dipoles induced in the two particles of the unit cell.

Although we have already justified the achirality of the unit cell of the arrays under consideration using symmetry arguments, for the sake of completeness, we provide a direct proof of it in Figure 5. To that end, we calculate the extinction cross-section σ_{ext} of an individual unit cell of the array considered in Figure 4. The pair of particles, whose spatial arrangement is described by the inset schematics, is excited with a plane wave propagating along the z -axis. The red solid curve shows the results calculated with the CDM for RCP excitation, which are exactly identical to those obtained for LCP excitation (green dashed curve). To ensure that there is not a chiral response arising from higher-order modes not captured by the dipole approximation, we also plot, with blue solid and yellow dashed curves, the value of σ_{ext} obtained from a full numerical solution of Maxwell's equations using a

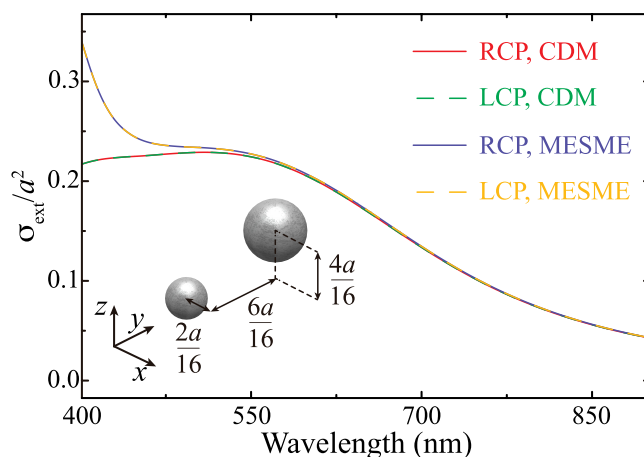


Figure 5. Achirality test for the unit cell. Extinction cross-section σ_{ext} of an individual unit cell of the 2.5-dimensional array considered in Figure 4 calculated with the CDM. The geometry of the particle dimer is described by the inset schematics. The system is excited with a plane wave propagating along the z -axis with either RCP (red solid curves) or LCP (green dashed curves) polarization. For comparison, the blue solid and yellow dashed curves show σ_{ext} obtained from a full numerical solution of Maxwell's equation using the MESME approach.

multiple elastic scattering of multipole expansions (MESME) approach.^{84,85} These spectra show no chiral response and also demonstrate that the accuracy of the CDM is excellent for wavelengths $\lambda \geq 550$ nm, which includes the spectral range considered in this work. Below 550 nm, the contribution of the quadrupole modes of the particles becomes more significant.

Up until now, we have analyzed square arrays with period $a = 800$ nm built from particles with diameters $D_1 = 120$ nm and $D_2 = 190$ nm. To prove that all of the results described in this work are general and not just a consequence of the particular choice of parameters, we perform an analysis of the response of arrays with different values of a and D_2 in Figure 6. In particular, as indicated by the legend, Figure 6(a) shows the transmittance spectra for arrays with $a = 750$ – 850 nm and $D_2 = 180$ – 200 nm, under RCP (solid curves) and LCP (dashed curves) excitation. In all cases, $D_1 = 120$ nm and $\mathbf{r}_2 = (2a, 6a, 4a)/16$. Analyzing these results, we observe that the increase in the period or the decrease in the diameter of particle 2 results in the lattice resonance moving closer to the Rayleigh anomaly and narrowing its lineshape. These behaviors, which have been described in previous works,^{9,76} are associated with the increase of the collective nature of the lattice resonance.^{17,30} A similar trend is observed for the dissymmetry factor displayed in Figure 6(b). Moreover, the increase in the ratio a/D_2 leads to larger values of g , practically reaching the theoretical limit of -2 . For instance, $g \sim -1.98$ for the array with $a = 850$ nm and $D_2 = 180$ nm. Therefore, these results confirm that the behavior described in this work is general, thus establishing a mechanism to achieve chiral lattice resonances based on the use of 2.5-dimensional arrays.

CONCLUSIONS

In summary, we have explored the optical response of 2.5-dimensional bipartite arrays made from the combination of a two-dimensional square lattice with a three-dimensional achiral unit cell. Using a rigorous approach based on the coupled dipole model, we have shown that these systems can sustain

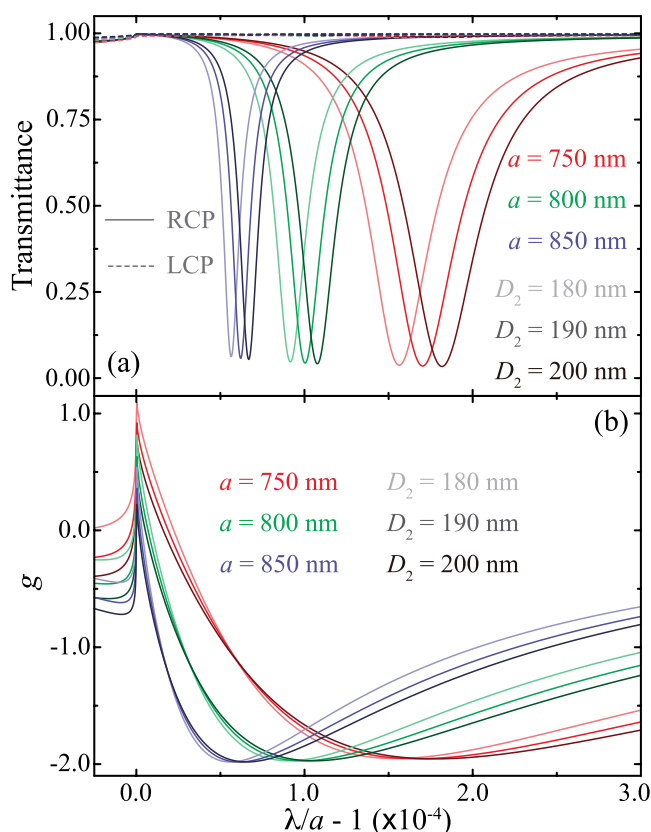


Figure 6. Effect of the array period and particle size on the chiral response of the array. (a) Transmittance for arrays with different values of a (see color code) and D_2 (see shade code) under RCP (solid curves) and LCP (dashed curves) excitation. In all cases, $D_1 = 120$ nm and $\mathbf{r}_2 = (2a, 6a, 4a)/16$. (b) Dissymmetry factor for the arrays analyzed in panel (a).

lattice resonances with almost perfect chiral behavior. The necessary conditions to obtain such resonances are (i) the use of two particles of different sizes and (ii) the placement of one of them in a low-symmetry position outside of the plane of the array. To illustrate this approach, we have analyzed the response of an array of period $a = 800$ nm, made of a unit cell with two silver particles of diameters $D_1 = 120$ nm and $D_2 = 190$ nm placed at positions $\mathbf{r}_1 = (0, 0, 0)$ and $\mathbf{r}_2 = (2a, 6a, 4a)/16$. Under RCP excitation, this system supports a strong chiral lattice resonance, which results in $T \approx 0.05$, $R \approx 0.63$, and $A \approx 0.32$ at the resonance peak, with a quality factor of 4×10^4 . In contrast, it displays a negligible response for LCP excitation, with $T \approx 0.99$. In consequence, the array behaves as a nearly perfect circular dichroic filter and a chiral mirror, which preserves the handedness of the reflected light. Importantly, the chiral response of the arrays considered here does not arise from the response of individual constituents, as is the case for arrays made of chiral structures, but rather from their collective interaction. Indeed, by decomposing the dipole moments induced in the particles, we have shown that the chiral response is the result of the constructive and destructive interference between the electric and magnetic dipoles induced in the two particles composing the unit cell.

For simplicity, we have assumed that the 2.5-dimensional arrays are surrounded by vacuum, but our analysis can be readily extended to any other homogeneous dielectric environment, with the only expected change being a shift in

the spectral position of the lattice resonances. However, it is well known that any asymmetry between the media above and below the array can hinder the lattice resonances and, hence, potentially affect the chiral responses predicted here. Fortunately, prior works have studied this problem and proposed strategies to minimize its effects,⁸⁶ including the use of index-matching fluids.⁸⁷ On the other hand, the presence of an inhomogeneous environment helps to break the mirror symmetry with respect to the array plane, potentially leading to an increase of the chiral response of the arrays.⁵⁹

The fabrication of this type of arrays is within reach of current experimental technology, as demonstrated by recent works.^{88–90} For instance, few-layer three-dimensional plasmonic arrays have been obtained by self-assembly of silica-coated gold nanoparticles that, once infiltrated with an index-matching fluid, behave as arrays suspended in a homogeneous environment.⁸⁸ Moreover, arrays with unit cells made of metallic nanostructures with different heights, and located at different planes, have been fabricated using an approach based on the binary-pore anodic aluminum oxide template technique.⁹⁰

The results of this work provide a strong theoretical understanding of the chiral response of the lattice resonances supported by structurally achiral 2.5-dimensional arrays, and therefore are expected to have far-reaching implications from an application point of view. For example, the dissymmetry in absorbance displayed by these systems upon excitation with light of different handedness could be exploited in thermoplasmonics applications³⁰ as a mechanism to control the heating of the array. Similarly, invoking reciprocity arguments, this same dissymmetry could be utilized to implement arrays emitting thermal radiation with nonvanishing optical helicity,⁹¹ or, if supplemented with a gain medium, sources of chiral light.⁹²

METHODS

Coupled Dipole Model. In order to describe the optical response of a 2.5-dimensional array built from a unit cell containing N particles, we use the coupled dipole model (CDM).^{4,5,9,78–80} This approach is justified when the particles are significantly smaller than both the wavelength of light and the period of the array. Within the CDM, and using Gaussian units, we model each of the particles as a point dipole $\mathbf{d}_{i,\mu} = (\mathbf{p}_{i,\mu}, \mathbf{m}_{i,\mu})$ with both electric $\mathbf{p}_{i,\mu}$ and magnetic $\mathbf{m}_{i,\mu}$ components. Furthermore, we use Greek indices to denote the unit cell to which the particle belongs and Latin ones to label the different particles within a unit cell. Accordingly, the position of any given particle in the array is specified by the sum of the vector $\mathbf{T}_\mu$, which indicates the location of the unit cell μ , and $\mathbf{r}_i$, which determines the position of particle i within the unit cell. It is important to note that, while $\mathbf{T}_\mu$ is contained in the xy -plane, $\mathbf{r}_i$ is a fully three-dimensional vector. The particles of the array are excited by a plane wave electromagnetic field $\mathbf{F}_{i,\mu} = (\mathbf{E}_{i,\mu}, \mathbf{H}_{i,\mu})$ with wavevector $\mathbf{k}$ and wavenumber $k = 2\pi/\lambda$. The components of the field are given by $\mathbf{E}_{i,\mu} = \mathbf{e}_0 e^{i\mathbf{k}_\parallel \mathbf{T}_\mu} e^{i\mathbf{k}_\parallel \mathbf{r}_i}$ and $\mathbf{H}_{i,\mu} = \mathbf{h}_0 e^{i\mathbf{k}_\parallel \mathbf{T}_\mu} e^{i\mathbf{k}_\parallel \mathbf{r}_i}$, with $\mathbf{e}_0$ and $\mathbf{h}_0$ being, respectively, the unit vectors of the electric and magnetic field components, and $\mathbf{k}_\parallel$ the projection of $\mathbf{k}$ onto the xy -plane. The dipole induced in an arbitrary particle of the array satisfies

$$\begin{pmatrix} \mathbf{p}_{i,\mu} \\ \mathbf{m}_{i,\mu} \end{pmatrix} = \begin{pmatrix} \alpha_i^{\text{EE}} & \alpha_i^{\text{EM}} \\ \alpha_i^{\text{ME}} & \alpha_i^{\text{MM}} \end{pmatrix} \begin{pmatrix} \mathbf{E}_{i,\mu} \\ \mathbf{H}_{i,\mu} \end{pmatrix} + \sum_{\nu} \sum_{j=1}^N \begin{pmatrix} \mathcal{G}_{ij,\mu\nu}^{\text{EE}} & \mathcal{G}_{ij,\mu\nu}^{\text{EM}} \\ \mathcal{G}_{ij,\mu\nu}^{\text{ME}} & \mathcal{G}_{ij,\mu\nu}^{\text{MM}} \end{pmatrix} \begin{pmatrix} \mathbf{p}_{j,\nu} \\ \mathbf{m}_{j,\nu} \end{pmatrix} \quad (1)$$

where EE, EM, ME, and MM stand for the electric–electric, electric–magnetic, magnetic–electric, and magnetic–magnetic components, respectively, and the prime in the first summation indicates that the terms $\nu = \mu$ are excluded from it when $i = j$ because a dipole does not interact with itself. Furthermore, α_i^{ζ} represent the different components of the polarizability tensor of the particles, while $\mathcal{G}_{ij,\mu\nu}^{\zeta}$ are the components of the dipole–dipole interaction tensor. The latter are defined as

$$\mathbf{G}_{ij,\mu\nu}^{\text{EE}} = \mathbf{G}_{ij,\mu\nu}^{\text{MM}} = [k^2 \mathcal{I}_{3 \times 3} + \nabla \nabla] \frac{e^{ik|\mathbf{T}_{\mu} + \mathbf{r}_i - \mathbf{T}_{\nu} - \mathbf{r}_j|}}{|\mathbf{T}_{\mu} + \mathbf{r}_i - \mathbf{T}_{\nu} - \mathbf{r}_j|}$$

and

$$\mathbf{G}_{ij,\mu\nu}^{\text{EM}} = -\mathbf{G}_{ij,\mu\nu}^{\text{ME}} = ik \nabla \times \frac{e^{ik|\mathbf{T}_{\mu} + \mathbf{r}_i - \mathbf{T}_{\nu} - \mathbf{r}_j|}}{|\mathbf{T}_{\mu} + \mathbf{r}_i - \mathbf{T}_{\nu} - \mathbf{r}_j|}$$

with $\mathcal{I}_{3 \times 3}$ being the 3×3 identity matrix. Exploiting the periodicity of the array, we can seek solutions of Equation 1 in the form of Bloch waves $\mathbf{d}_{i,\mu} = \mathbf{d}_i e^{ik_{\parallel} \mathbf{T}_{\mu}} = (\mathbf{p}_i, \mathbf{m}_i) e^{ik_{\parallel} \mathbf{T}_{\mu}}$. By doing so, we get

$$\begin{pmatrix} \mathbf{p}_i \\ \mathbf{m}_i \end{pmatrix} = \begin{pmatrix} \alpha_i^{\text{EE}} & \alpha_i^{\text{EM}} \\ \alpha_i^{\text{ME}} & \alpha_i^{\text{MM}} \end{pmatrix} \begin{pmatrix} \mathbf{E}_i \\ \mathbf{H}_i \end{pmatrix} + \sum_{j=1}^N \begin{pmatrix} \mathcal{G}_{ij}^{\text{EE}} & \mathcal{G}_{ij}^{\text{EM}} \\ -\mathcal{G}_{ij}^{\text{EM}} & \mathcal{G}_{ij}^{\text{EE}} \end{pmatrix} \begin{pmatrix} \mathbf{p}_j \\ \mathbf{m}_j \end{pmatrix}$$

where $\mathcal{G}_{ij}^{\zeta} = \sum_{\nu} \mathbf{G}_{ij,\nu 0}^{\zeta} e^{-ik_{\parallel} \mathbf{T}_{\nu}}$ represent the Fourier transform of the different components of the dipole–dipole interaction tensor, usually referred to as the lattice sum. The equation above can be solved as

$$\begin{pmatrix} \mathbf{p}_i \\ \mathbf{m}_i \end{pmatrix} = \sum_{j=1}^N \begin{pmatrix} \mathcal{A}_{ij}^{\text{EE}} & \mathcal{A}_{ij}^{\text{EM}} \\ \mathcal{A}_{ij}^{\text{ME}} & \mathcal{A}_{ij}^{\text{MM}} \end{pmatrix} \begin{pmatrix} \mathbf{E}_j \\ \mathbf{H}_j \end{pmatrix}$$

with $\mathcal{A}_{ij}^{\zeta}$ being the ζ component of the ij -th block of the $6N \times 6N$ tensor $\mathcal{A} = [\alpha^{-1} - \mathcal{G}]^{-1}$ that contains the effective polarizability of the array. This quantity encodes the optical response of the array, which stems from the interplay between the individual response of the particles described by α and their mutual interactions determined by the lattice sum $\mathcal{G}$.

We can particularize the result shown above for the bipartite arrays considered in this work:

$$\begin{pmatrix} \mathbf{p}_1 \\ \mathbf{m}_1 \\ \mathbf{p}_2 \\ \mathbf{m}_2 \end{pmatrix} = \begin{pmatrix} \mathcal{A}_{11}^{\text{EE}} & \mathcal{A}_{11}^{\text{EM}} & \mathcal{A}_{12}^{\text{EE}} & \mathcal{A}_{12}^{\text{EM}} \\ \mathcal{A}_{11}^{\text{ME}} & \mathcal{A}_{11}^{\text{MM}} & \mathcal{A}_{12}^{\text{ME}} & \mathcal{A}_{12}^{\text{MM}} \\ \mathcal{A}_{21}^{\text{EE}} & \mathcal{A}_{21}^{\text{EM}} & \mathcal{A}_{22}^{\text{EE}} & \mathcal{A}_{22}^{\text{EM}} \\ \mathcal{A}_{21}^{\text{ME}} & \mathcal{A}_{21}^{\text{MM}} & \mathcal{A}_{22}^{\text{ME}} & \mathcal{A}_{22}^{\text{MM}} \end{pmatrix} \begin{pmatrix} \mathbf{E}_1 \\ \mathbf{H}_1 \\ \mathbf{E}_2 \\ \mathbf{H}_2 \end{pmatrix} \quad (2)$$

Notice that, since we consider spherical particles made of silver, $\alpha_i^{\text{EM}} = \alpha_i^{\text{ME}} = 0$, while α_i^{EE} and α_i^{MM} are diagonal and can be computed from the dipolar Mie scattering coefficient⁸⁵ (see Figure S3 of the Supporting Information).

Once the induced electric and magnetic dipoles are known, we can calculate the electric field created by the array at an

arbitrary point $\mathbf{r} = (\mathbf{R}, z)$ as $\mathbf{E}_{\text{rad}} = \sum_{\mathbf{q}} \mathbf{E}_{\mathbf{q}}$ where $\mathbf{q}$ are the reciprocal lattice vectors, and

$$\mathbf{E}_{\mathbf{q}} = \frac{2\pi i}{S} \sum_{i=1}^N [k^2 \mathbf{p}_i - \mathbf{k}_{\mathbf{q}}(\mathbf{k}_{\mathbf{q}} \cdot \mathbf{p}_i) - k \mathbf{k}_{\mathbf{q}} \times \mathbf{m}_i] \frac{e^{i(\mathbf{k}_{\parallel} + \mathbf{q}) \cdot (\mathbf{R} - \mathbf{R}_i)}}{k_{z\mathbf{q}}} e^{ik_{z\mathbf{q}}|z - z_i|}$$

where S is the area of the unit cell, $\mathbf{r}_i = (\mathbf{R}_i, z_i)$ are the positions of the particles in the unit cell, $\mathbf{k}_{\mathbf{q}} = (\mathbf{k}_{\parallel} + \mathbf{q}, \pm k_{z\mathbf{q}})$, and $k_{z\mathbf{q}} = \sqrt{k^2 - |\mathbf{k}_{\parallel} + \mathbf{q}|^2}$. Notice that the upper (lower) signs apply for z larger (smaller) than z_i . Finally, we calculate the transmittance, reflectance, and absorbance of the array from the power transmitted, reflected, and absorbed, respectively, normalized to the incident power.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsp Photonics.3c00369>.

Benchmark of the coupled dipole model against full numerical solutions of Maxwell's equations. Analysis of the role of the magnetic response in the chirality of the arrays. Plot of the electric and magnetic polarizabilities of the nanoparticles. (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Alejandro Manjavacas – Instituto de Óptica (IO-CSIC), Consejo Superior de Investigaciones Científicas, 28006 Madrid, Spain; Department of Physics and Astronomy, University of New Mexico, Albuquerque, New Mexico 87106, United States; orcid.org/0000-0002-2379-1242; Email: a.manjavacas@csic.es

Authors

Luis Cerdán – Instituto de Óptica (IO-CSIC), Consejo Superior de Investigaciones Científicas, 28006 Madrid, Spain; orcid.org/0000-0002-7174-2453

Lauren Zundel – Department of Physics and Astronomy, University of New Mexico, Albuquerque, New Mexico 87106, United States; orcid.org/0000-0003-1850-5210

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acsp Photonics.3c00369>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was sponsored by a 2022 Leonardo Grant for Researchers in Physics from the BBVA Foundation. The authors also acknowledge support from Grant No. PID2019-109502GA-I00 funded by MCIN/AEI/10.13039/501100011033 as well as the U.S. National Science Foundation (Grant No. DMR-1941680). L.Z. acknowledges support from the Department of Energy Computational Science Graduate Fellowship (Grant No. DE-SC0020347).

■ REFERENCES

- (1) Kravets, V. G.; Kabashin, A. V.; Barnes, W. L.; Grigorenko, A. N. Plasmonic surface lattice resonances: A review of properties and applications. *Chem. Rev.* **2018**, *118*, 5912–5951.

- (2) Wang, W.; Ramezani, M.; Väkeväinen, A. I.; Törmä, P.; Gómez Rivas, J.; Odom, T. W. The rich photonic world of plasmonic nanoparticle arrays. *Mater. Today* **2018**, *21*, 303–314.
- (3) Utyushev, A. D.; Zakomirnyi, V. I.; Rasskazov, I. L. Collective lattice resonances: Plasmonics and beyond. *Rev. Phys.* **2021**, *6*, 100051.
- (4) Baur, S.; Sanders, S.; Manjavacas, A. Hybridization of lattice resonances. *ACS Nano* **2018**, *12*, 1618–1629.
- (5) Cherqui, C.; Bourgeois, M. R.; Wang, D.; Schatz, G. C. Plasmonic surface lattice resonances: Theory and computation. *Acc. Chem. Res.* **2019**, *52*, 2548–2558.
- (6) Adato, R.; Yanik, A. A.; Wu, C.-H.; Shvets, G.; Altug, H. Radiative engineering of plasmon lifetimes in embedded nanoantenna arrays. *Opt. Express* **2010**, *18*, 4526–4537.
- (7) Zakomirnyi, V. I.; Rasskazov, I. L.; Gerasimov, V. S.; Ershov, A. E.; Polyutov, S. P.; Karpov, S. V. Refractory titanium nitride two-dimensional structures with extremely narrow surface lattice resonances at telecommunication wavelengths. *Appl. Phys. Lett.* **2017**, *111*, 123107.
- (8) Khlopov, D.; Laux, F.; Wardley, W. P.; Martin, J.; Wurtz, G. A.; Plain, J.; Bonod, N.; Zayats, A. V.; Dickson, W.; Gérard, D. Lattice modes and plasmonic linewidth engineering in gold and aluminum nanoparticle arrays. *J. Opt. Soc. Am. B* **2017**, *34*, 691–700.
- (9) Cuartero-González, A.; Sanders, S.; Zundel, L.; Fernández-Domínguez, A. I.; Manjavacas, A. Super- and subradiant lattice resonances in bipartite nanoparticle arrays. *ACS Nano* **2020**, *14*, 11876.
- (10) Le-Van, Q.; Zoethout, E.; Geluk, E.-J.; Ramezani, M.; Berghuis, M.; Gómez Rivas, J. Enhanced quality factors of surface lattice resonances in plasmonic arrays of nanoparticles. *Adv. Opt. Mater.* **2019**, *7*, 1801451.
- (11) Deng, S.; Li, R.; Park, J.-E.; Guan, J.; Choo, P.; Hu, J.; Smeets, P. J. M.; Odom, T. W. Ultranarrow plasmon resonances from annealed nanoparticle lattices. *Proc. Natl. Acad. Sci. U. S. A.* **2020**, *117*, 23380–23384.
- (12) Bin-Alam, M. S.; Reshef, O.; Mamchur, Y.; Alam, M. Z.; Carlow, G.; Upham, J.; Sullivan, B. T.; Ménard, J.-M.; Huttunen, M. J.; Boyd, R. W.; Dolgaleva, K. Ultra-high-Q resonances in plasmonic metasurfaces. *Nat. Commun.* **2021**, *12*, 974.
- (13) Zundel, L.; May, A.; Manjavacas, A. Lattice resonances induced by periodic vacancies in arrays of nanoparticles. *ACS Photonics* **2021**, *8*, 360–368.
- (14) Zou, S.; Schatz, G. C. Silver nanoparticle array structures that produce giant enhancements in electromagnetic fields. *Chem. Phys. Lett.* **2005**, *403*, 62–67.
- (15) Nikitin, A. G.; Kabashin, A. V.; Dallaporta, H. Plasmonic resonances in diffractive arrays of gold nanoantennas: near and far field effects. *Opt. Express* **2012**, *20*, 27941–27952.
- (16) Manjavacas, A.; Zundel, L.; Sanders, S. Analysis of the limits of the near-field produced by nanoparticle arrays. *ACS Nano* **2019**, *13*, 10682–10693.
- (17) Zundel, L.; Deop-Ruano, J. R.; Martínez-Herrero, R.; Manjavacas, A. Lattice resonances excited by finite-width light beams. *ACS Omega* **2022**, *7*, 31431–31441.
- (18) Bai, Y.-T.; Zheng, H.-Y.; Zhang, Q.; Liu, S.-D. Hybridized magnetic lattice resonances for narrowband perfect absorption. *Opt. Laser Technol.* **2022**, *156*, 108460.
- (19) Adato, R.; Yanik, A. A.; Amsden, J. J.; Kaplan, D. L.; Omenetto, F. G.; Hong, M. K.; Erramilli, S.; Altug, H. Ultra-sensitive vibrational spectroscopy of protein monolayers with plasmonic nanoantenna arrays. *Proc. Natl. Acad. Sci. U. S. A.* **2009**, *106*, 19227–19232.
- (20) Thackray, B. D.; Kravets, V. G.; Schedin, F.; Auton, G.; Thomas, P. A.; Grigorenko, A. N. Narrow collective plasmon resonances in nanostructure arrays observed at normal light incidence for simplified sensing in asymmetric air and water environments. *ACS Photonics* **2014**, *1*, 1116–1126.
- (21) Danilov, A.; Tselikov, G.; Wu, F.; Kravets, V. G.; Ozerov, I.; Bedu, F.; Grigorenko, A. N.; Kabashin, A. V. Ultra-narrow surface lattice resonances in plasmonic metamaterial arrays for biosensing applications. *Biosens. Bioelectron.* **2018**, *104*, 102–112.
- (22) Kristensen, A.; Yang, J. K. W.; Bozhevolnyi, S. I.; Link, S.; Nordlander, P.; Halas, N. J.; Mortensen, N. A. Plasmonic colour generation. *Nat. Rev. Mater.* **2017**, *2*, 16088.
- (23) Esposito, M.; Todisco, F.; Bakhti, S.; Passaseo, A.; Tarantini, I.; Cuscunà, M.; Destouches, N.; Tasco, V. Symmetry breaking in oligomer surface plasmon lattice resonances. *Nano Lett.* **2019**, *19*, 1922–1930.
- (24) Lozano, G.; Louwers, D. J.; Rodríguez, S. R. K.; Murai, S.; Jansen, O. T. A.; Verschuuren, M. A.; Gómez Rivas, J. Plasmonics for solid-state lighting: enhanced excitation and directional emission of highly efficient light sources. *Light Sci. Appl.* **2013**, *2*, No. e66.
- (25) Zhou, W.; Dridi, M.; Suh, J. Y.; Kim, C. H.; Co, D. T.; Wasielewski, M. R.; Schatz, G. C.; Odom, T. W. Lasing action in strongly coupled plasmonic nanocavity arrays. *Nat. Nanotechnol.* **2013**, *8*, 506–511.
- (26) Schokker, A. H.; Koenderink, A. F. Lasing at the band edges of plasmonic lattices. *Phys. Rev. B* **2014**, *90*, 155452.
- (27) Guo, R.; Nečada, M.; Hakala, T. K.; Väkeväinen, A. I.; Törmä, P. Lasing at K points of a honeycomb plasmonic lattice. *Phys. Rev. Lett.* **2019**, *122*, 013901.
- (28) Vaskin, A.; Kolkowski, R.; Koenderink, F. A.; Staude, I. Light-emitting metasurfaces. *Nanophotonics* **2019**, *8*, 1151.
- (29) Bourgeois, M. R.; Rossi, A. W.; Khorasani, S.; Masiello, D. J. Optical control over thermal distributions in topologically trivial and non-trivial plasmon lattices. *ACS Photonics* **2022**, *9*, 3656–3667.
- (30) Zundel, L.; Malone, K.; Cerdán, L.; Martínez-Herrero, R.; Manjavacas, A. Lattice resonances for thermoplasmonics. *ACS Photonics* **2023**, *10*, 274–282.
- (31) Väkeväinen, A. I.; Moerland, R. J.; Rekola, H. T.; Eskelinen, A.-P.; Martikainen, J.-P.; Kim, D.-H.; Törmä, P. Plasmonic surface lattice resonances at the strong coupling regime. *Nano Lett.* **2014**, *14*, 1721–1727.
- (32) Rodríguez, S. R. K.; Feist, J.; Verschuuren, M. A.; Garcia Vidal, F. J.; Gómez Rivas, J. Thermalization and cooling of plasmon-exciton polaritons: Towards quantum condensation. *Phys. Rev. Lett.* **2013**, *111*, 166802.
- (33) Hakala, T. K.; Moilanen, A. J.; Väkeväinen, A. I.; Guo, R.; Martikainen, J.-P.; Daskalakis, K. S.; Rekola, H. T.; Julku, A.; Törmä, P. Bose–Einstein condensation in a plasmonic lattice. *Nat. Phys.* **2018**, *14*, 739–744.
- (34) Yadav, R. K.; Bourgeois, M. R.; Cherqui, C.; Juarez, X. G.; Wang, W.; Odom, T. W.; Schatz, G. C.; Basu, J. K. Room temperature weak-to-strong coupling and the emergence of collective emission from quantum dots coupled to plasmonic arrays. *ACS Nano* **2020**, *14*, 7347–7357.
- (35) Boddetti, A. K.; Guan, J.; Sentz, T.; Juarez, X.; Newman, W.; Cortes, C.; Odom, T. W.; Jacob, Z. Long-range dipole-dipole interactions in a plasmonic lattice. *Nano Lett.* **2022**, *22*, 22–28.
- (36) Zundel, L.; Cuartero-González, A.; Sanders, S.; Fernández-Domínguez, A. I.; Manjavacas, A. Green tensor analysis of lattice resonances in periodic arrays of nanoparticles. *ACS Photonics* **2022**, *9*, 540–550.
- (37) Cotrufo, M.; Osorio, C. I.; Koenderink, A. F. Spin-dependent emission from arrays of planar chiral nanoantennas due to lattice and localized plasmon resonances. *ACS Nano* **2016**, *10*, 3389–3397.
- (38) Mattioli, F.; et al. Plasmonic superchiral lattice resonances in the mid-infrared. *ACS Photonics* **2020**, *7*, 2676–2681.
- (39) Liu, S.-D.; Liu, J.-Y.; Cao, Z.; Fan, J.-L.; Lei, D. Dynamic tuning of enhanced intrinsic circular dichroism in plasmonic stereo-metamolecule array with surface lattice resonance. *Nanophotonics* **2020**, *9*, 3419–3434.
- (40) Goerlitz, E. S. A.; Zapata-Herrera, M.; Ponomareva, E.; Feller, D.; García-Etxarri, A.; Karg, M.; Aizpurua, J.; Vogel, N. Molecular-induced chirality transfer to plasmonic lattice modes. *arXiv*, January 19, 2023. arXiv:2301.06761, ver. 2.
- (41) Wagnière, G. H. *On chirality and the universal asymmetry*; VCH: Wiley-VCH: Zürich, Switzerland, 2007.

- (42) Meierhenrich, U. *Amino acids and the asymmetry of life: Caught in the act of formation*; Springer Science & Business Media: Berlin/Heidelberg, Germany, 2008.
- (43) Nguyen, L. A.; He, H.; Pham-Huy, C. Chiral drugs: An overview. *Int. J. Biomed. Sci.* **2006**, *2*, 85–100.
- (44) Smith, S. W. Chiral toxicology: It's the same thing...only different. *Toxicol. Sci.* **2009**, *110*, 4–30.
- (45) Kim, J. H.; Scialli, R. Thalidomide: The tragedy of birth defects and the effective treatment of disease. *Toxicol. Sci.* **2011**, *122*, 1–6.
- (46) García-Guirado, J.; Svedendahl, M.; Puigdollers, J.; Quidant, R. Enantiomer-selective molecular sensing using racemic nanoplasmonic arrays. *Nano Lett.* **2018**, *18*, 6279–6285.
- (47) Liu, T.; Besteiro, L. V.; Liedl, T.; Correa-Duarte, M. A.; Wang, Z.; Govorov, A. O. Chiral plasmonic nanocrystals for generation of hot electrons: Toward polarization-sensitive photochemistry. *Nano Lett.* **2019**, *19*, 1395–1407.
- (48) Paiva-Marques, W. A.; Reyes Gómez, F.; Oliveira, O. N.; Mejía-Salazar, J. R. Chiral plasmonics and their potential for point-of-care biosensing applications. *Sensors* **2020**, *20*, 944.
- (49) Goerlitzer, E. S. A.; Puri, A. S.; Moses, J. J.; Poulidakos, L. V.; Vogel, N. The beginner's guide to chiral plasmonics: Mostly harmless theory and the design of large-area substrates. *Adv. Opt. Mater.* **2021**, *9*, 2100378.
- (50) Warning, L. A.; Miasdashti, A. R.; McCarthy, L. A.; Zhang, Q.; Landes, C. F.; Link, S. Nanophotonic approaches for chirality sensing. *ACS Nano* **2021**, *15*, 15538–15566.
- (51) Kim, R. M.; et al. Enantioselective sensing by collective circular dichroism. *Nature* **2022**, *612*, 470–476.
- (52) Mendoza-Carreño, J.; Molet, P.; Otero-Martínez, C.; Alonso, M. I.; Polavarapu, L.; Mihi, A. Nanoimprinted 2D-chiral perovskite nanocrystal metasurfaces for circularly polarized photoluminescence. *Adv. Mater.* **2023**, *35*, 2210477.
- (53) Collins, J. T.; Kuppe, C.; Hooper, D. C.; Sibilia, C.; Centini, M.; Valev, V. K. Chirality and chiroptical effects in metal nanostructures: Fundamentals and current trends. *Adv. Opt. Mater.* **2017**, *5*, 1700182.
- (54) Mun, J.; Kim, M.; Yang, Y.; Badloe, T.; Ni, J.; Chen, Y.; Qiu, C.-W.; Rho, J. Electromagnetic chirality: From fundamentals to nontraditional chiroptical phenomena. *Light Sci. Appl.* **2020**, *9*, 139.
- (55) Ávalos-Ovando, O.; et al. Chiral bioinspired plasmonics: A paradigm shift for optical activity and photochemistry. *ACS Photonics* **2022**, *9*, 2219–2236.
- (56) Goerlitzer, E. S. A.; Mohammadi, R.; Nechayev, S.; Volk, K.; Rey, M.; Banzer, P.; Karg, M.; Vogel, N. Chiral surface lattice resonances. *Adv. Mater.* **2020**, *32*, 2001330.
- (57) Ji, C.-Y.; Chen, S.; Han, Y.; Liu, X.; Liu, J.; Li, J.; Yao, Y. Artificial propeller chirality and counterintuitive reversal of circular dichroism in twisted meta-molecules. *Nano Lett.* **2021**, *21*, 6828–6834.
- (58) Li, X.; Ji, C.-Y.; Chen, S.; Han, Y.; Liu, J.; Li, J. Phase enabled circular dichroism reversal in twisted bi-chiral propeller metamolecule arrays. *Adv. Opt. Mater.* **2021**, *9*, 2101191.
- (59) Movsesyan, A.; Besteiro, L. V.; Kong, X.-T.; Wang, Z.; Govorov, A. O. Engineering strongly chiral plasmonic lattices with achiral unit cells for sensing and photodetection. *Adv. Opt. Mater.* **2022**, *10*, 2101943.
- (60) Esposito, M.; Manoccio, M.; Leo, A.; Cuscunà, M.; Sun, Y.; Ageev, E.; Zuev, D.; Benedetti, A.; Tarantini, I.; Passaseo, A.; Tasco, V. 3D chiral metacrystals. *Adv. Funct. Mater.* **2022**, *32*, 2109258.
- (61) Manoccio, M.; Tasco, V.; Todisco, F.; Passaseo, A.; Cuscunà, M.; Tarantini, I.; Gigli, G.; Esposito, M. Surface lattice resonances in 3D chiral metacrystals for plasmonic sensing. *Adv. Sci.* **2023**, *10*, 2206930.
- (62) Huang, S.; Yang, X.; Liang, X.; Wu, X.; Yang, C.; Du, J.; Hou, Y. Engineering a strong and stable ultraviolet chiroptical effect in a large-area chiral plasmonic shell. *Opt. Express* **2022**, *30*, 31486–31497.
- (63) Qiao, S.; Liang, Q.; Zhang, X.; Liu, X.; Feng, S.; Ji, C.-Y.; Guo, H.; Li, J. Flexible engineering of circular dichroism enabled by chiral surface lattice resonances. *APL Photon* **2022**, *7*, 116104.
- (64) Romashkina, A. M.; Novikov, V. B.; Murzina, T. V. Collective lattice and plasmonic resonances in the enhancement of circular dichroism in disk-rod metasurface. *J. Appl. Phys.* **2023**, *133*, 043103.
- (65) Zhao, S.; Shao, L.; Wang, J.; Lin, H.-Q.; Zhang, W. Chirality-selective transparency induced by lattice resonance in bilayer metasurfaces. *Photon. Res.* **2021**, *9*, 484–493.
- (66) Bai, Y.; Li, Y.; Ikram, M.; Ren, Y.; Xu, Y.; Wang, Y.; Huo, Y.; Zhang, Z. Circular dichroism induced by the coupling between surface plasmon polaritons and localized surface plasmon resonances in a double-layer complementary nanostructure. *J. Phys. Chem. C* **2022**, *126*, 10159–10166.
- (67) Apurv Chaitanya, N.; Butt, M. A. T.; Reshef, O.; Boyd, R. W.; Banzer, P.; De Leon, I. Lattice-plasmon-induced asymmetric transmission in two-dimensional chiral arrays. *APL Photonics* **2022**, *7*, 016105.
- (68) Plum, E.; Liu, X.-X.; Fedotov, V. A.; Chen, Y.; Tsai, D. P.; Zheludev, N. I. Metamaterials: Optical activity without chirality. *Phys. Rev. Lett.* **2009**, *102*, 113902.
- (69) Plum, E.; Fedotov, V. A.; Zheludev, N. I. Extrinsic electromagnetic chirality in metamaterials. *J. Opt. Soc. Am.* **2009**, *11*, 074009.
- (70) De Leon, I.; Horton, M. J.; Schulz, S. A.; Upham, J.; Banzer, P.; Boyd, R. W. Strong, spectrally-tunable chirality in diffractive metasurfaces. *Sci. Rep.* **2015**, *5*, 13034.
- (71) Chen, S.; Reineke, B.; Li, G.; Zentgraf, T.; Zhang, S. Strong nonlinear optical activity induced by lattice surface modes on plasmonic metasurface. *Nano Lett.* **2019**, *19*, 6278–6283.
- (72) Petronijevic, E.; Belardini, A.; Cesca, T.; Scian, C.; Mattei, G.; Sibilia, C. Rich near-infrared chiral behavior in diffractive metasurfaces. *Phys. Rev. Applied* **2021**, *16*, 014003.
- (73) Yang, H. U.; D'Archangel, J.; Sundheimer, M. L.; Tucker, E.; Boreman, G. D.; Raschke, M. B. Optical dielectric function of silver. *Phys. Rev. B* **2015**, *91*, 235137.
- (74) Fan, Z.; Govorov, A. O. Plasmonic circular dichroism of chiral metal nanoparticle assemblies. *Nano Lett.* **2010**, *10*, 2580–2587.
- (75) Martens, K.; Funck, T.; Santiago, E. Y.; Govorov, A. O.; Burger, S.; Liedl, T. Onset of chirality in plasmonic meta-molecules and dielectric coupling. *ACS Nano* **2022**, *16*, 16143–16149.
- (76) Deop-Ruano, J. R.; Sanders, S.; Alabastri, A.; Kort-Kamp, W. J. M.; Dalvit, D. A. R.; Manjavacas, A. Optical response of periodic arrays of graphene nanodisks. *Phys. Rev. Applied* **2022**, *18*, 044071.
- (77) García de Abajo, F. J. Colloquium: Light scattering by particle and hole arrays. *Rev. Mod. Phys.* **2007**, *79*, 1267–1290.
- (78) Teperik, T. V.; Degiron, A. Design strategies to tailor the narrow plasmon-photonic resonances in arrays of metallic nanoparticles. *Phys. Rev. B* **2012**, *86*, 245425.
- (79) Kolkowski, R.; Koenderink, A. F. Lattice resonances in optical metasurfaces with gain and loss. *Proc. IEEE* **2020**, *108*, 795–818.
- (80) Zundel, L.; Manjavacas, A. Finite-size effects on periodic arrays of nanostructures. *J. Phys.: Photonics* **2019**, *1*, 015004.
- (81) Beutel, D.; Fernandez-Corbaton, I.; Rockstuhl, C. Unified lattice sums accommodating multiple sublattices for solutions of the Helmholtz equation in two and three dimensions. *Phys. Rev. A* **2023**, *107*, 013508.
- (82) Sánchez-Carnerero, E. M.; Agarrabeitia, A. R.; Moreno, F.; Maroto, B. L.; Muller, G.; Ortiz, M. J.; de la Moya, S. Circularly polarized luminescence from simple organic molecules. *Chem. Eur. J.* **2015**, *21*, 13488–13500.
- (83) Mao, L.; Liu, K.; Zhang, S.; Cao, T. Extrinsic 2D-chiral metamirror in near-infrared region. *ACS Photonics* **2020**, *7*, 375–383.
- (84) García de Abajo, F. J. Interaction of radiation and fast electrons with clusters of dielectrics: A multiple scattering approach. *Phys. Rev. Lett.* **1999**, *82*, 2776–2779.
- (85) García de Abajo, F. J. Multiple scattering of radiation in clusters of dielectrics. *Phys. Rev. B* **1999**, *60*, 6086–6102.
- (86) Auguie, B.; Bendaña, X. M.; Barnes, W. L.; García de Abajo, F. J. Diffractive arrays of gold nanoparticles near an interface: Critical role of the substrate. *Phys. Rev. B* **2010**, *82*, 155447.

(87) Volk, K.; Fitzgerald, J. P. S.; Karg, M. In-plane surface lattice and higher order resonances in self-assembled plasmonic monolayers: From substrate-supported to free-standing thin films. *ACS Appl. Mater. Interfaces* **2019**, *11*, 16096–16106.

(88) Hasegawa, M.; Watanabe, K.; Namigata, H.; Welling, T. A.; Suga, K.; Nagao, D. Surface lattice resonance in three-dimensional plasmonic arrays fabricated via self-assembly of silica-coated gold nanoparticles. *J. Colloid Interface Sci.* **2023**, *633*, 226–232.

(89) Kühner, L.; Wendisch, F. J.; Antonov, A. A.; Bürger, J.; Hüttenhofer, L.; Menezes, L. d. S.; Maier, S. A.; Gorkunov, M. V.; Kivshar, Y.; Tittl, A. Unlocking the out-of-plane dimension for photonic bound states in the continuum to achieve maximum optical chirality. *arXiv*, October 11, 2022, 2210.05339, ver. 1.

(90) Wang, Z.; Sun, J.; Li, J.; Wang, L.; Li, Z.; Zheng, X.; Wen, L. Customizing 2.5D out-of-plane architectures for robust plasmonic bound-states-in-the-continuum metasurfaces. *Adv. Sci.* **2023**, *10*, 2206236.

(91) Wang, X.; Sentz, T.; Bharadwaj, S.; Ray, S. K.; Wang, Y.; Jiao, D.; Qi, L.; Jacob, Z. Observation of nonvanishing optical helicity in thermal radiation from symmetry-broken metasurfaces. *Sci. Adv.* **2023**, *9*, No. eade4203.

(92) Zhang, X.; Liu, Y.; Han, J.; Kivshar, Y.; Song, Q. Chiral emission from resonant metasurfaces. *Science* **2022**, *377*, 1215–1218.



CAS BIOFINDER DISCOVERY PLATFORM™

CAS BIOFINDER HELPS YOU FIND YOUR NEXT BREAKTHROUGH FASTER

Navigate pathways, targets, and
diseases with precision

Explore CAS BioFinder

